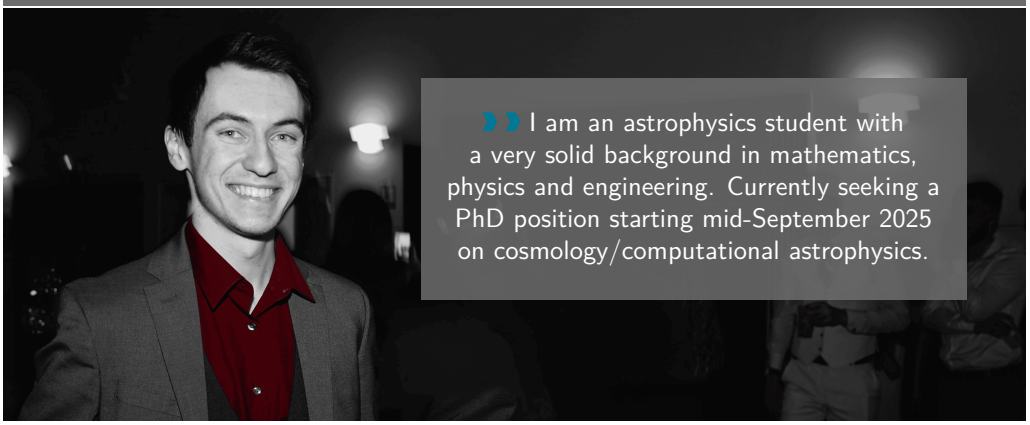
SAMUEL MESQUITA | CV



I am an astrophysics student with a very solid background in mathematics, physics and engineering. Currently seeking a PhD position starting mid-September 2025 on cosmology/computational astrophysics.

STATUS

Found an internship of 6-month starting February 2025 (cf. *Projects* section).
Looking for a PhD starting mid-September 2025 on cosmology/computational astrophysics.

EDUCATION

Astrophysics & Data Science Master 2 — UNISTRA - France, 2024 - 2025

- Study current astrophysical knowledge and methods: galaxy physics, stellar evolution, high-energy phenomena, virtual observatory tools and numerical simulations

QMat Excellence Scholarship — UNISTRA - France, 2024 - 2025

General Engineering Formation — Télécom Physique Strasbourg - France, 2022 - 2025

- Multidisciplinary curriculum which offers a strong approach to understanding diverse engineering and physics principles and applications

Preparatory Classes — Kerichen High School - Brest, France, 2020 - 2022

- Intensive program specialized in mathematics and physics, aimed at preparing students for entrance exams to prestigious engineering schools

PROJECTS

6-month internship — Astronomical Observatory of Strasbourg - France, 02-08/2025

- Implementation of a model of star & galaxy formation for the new hybrid simulation code *Dyablo* (C++) using proven semi-analytical formalisms of dark matter halo formation

3-month internship — DIFFER - Netherlands, 06 - 08/2024

- Machine learning exploration of ORR catalytic activity on metallic (110) surfaces
- Use of Python (*fairchem* & *pandas* packages) to discover new catalysts to efficiently store renewable energies. Focused on the Oxygen Reduction Reactions in a fuel cell.

Circumstellar disk — Astronomical Observatory of Strasbourg - France, 05/2024

- Use of Python (*amuse* package) to study the parameters influencing the capture of a satellite by a planet in a forming solar system with a circumstellar disk

Engineering project — Télécom Physique Strasbourg - France, 2023 - 2024

- Use of Python (*scipy* & *tkinter* packages) to develop a software to monitor the supply chain and compute statistical data for nuclear engineering, features a graphical user interface

1-month internship — Astronomical Observatory of Strasbourg - France, 06/2023

- Use of IDL to study the radio emission and star formation in cluster spiral galaxies

Competitive exams project — Kerichen High School - Brest, France, 2021-2022

- Use of Python (*numpy* & *matplotlib* packages) to identify optimal strategies to minimize oscillations in an earthquake-resistant building, adapted for diverse structural designs

INFORMATIONS

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LANGUAGES

- French (Native) ★★★★★
- English (C1) ★★★★★
- Spanish (A2) ★★
- German (A2) ★★

SKILLS

- Physics
- Mathematics
- Signal processing
- Computer science
- Statistical analysis
- Data manipulation

IT SKILLS

- Python
- C/C++
- LaTeX
- Unix
- Matlab
- Labview
- Comsol
- SQL/ADQL

INTERESTS

- Badminton
- Board games
- Video games